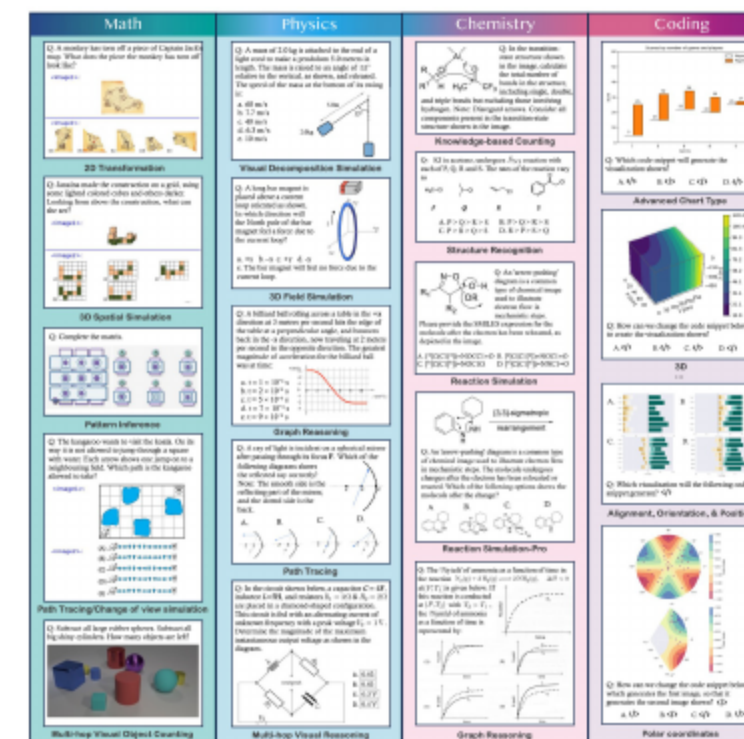


Can MLLMs Reason in Multimodality? EMMA Benchmark Overview

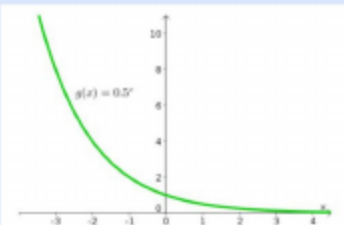
Yunzhuo Hao; Jiawei Gu; Huichen Will Wang; Linjie Li; Zhengyuan Yang; Lijuan Wang; Yu Cheng

01 Why Multimodal Reasoning Matters



- Human intelligence blends modalities: text supports abstraction while images support spatial insight
- Real tasks demand integration, from design to scientific analysis, not siloed skills
- For MLLMs, coordinating visual interpretation with language reasoning yields **more robust** solutions
- Multimodal reasoning reduces brittle shortcuts and encourages stepwise, grounded problem solving

02 From Text-Only Progress to a Multimodal Gap

Q: Is this an odd function? Yes or No  C: The function is $y = 0.5^x$... ☒ C+Q: No ☒ I+Q: No ☒	Q: A trip of the pupils to the zoo took 135 minutes. How many hours and minutes does it make?  C: The image contains a bus... ☒ C+Q: 2 h 15 min ☒ I+Q: 2 h 15 min ☒	Q: Chase wants to buy 4 kilograms of oval beads and 5 kilograms of star-shaped beads. How much will he spend? <table> <tr><td>oval beads</td><td>\$2/kilogram</td></tr> <tr><td>rectangular beads</td><td>\$3/kilogram</td></tr> <tr><td>star-shaped beads</td><td>\$2/kilogram</td></tr> <tr><td>spherical beads</td><td>\$2/kilogram</td></tr> <tr><td>heart-shaped beads</td><td>\$3/kilogram</td></tr> <tr><td>square beads</td><td>\$2/kilogram</td></tr> <tr><td>flower-shaped beads</td><td>\$2/kilogram</td></tr> </table> C: The image is a price list ... ☒ C+Q: \$18 ☒ I+Q: \$18 ☒	oval beads	\$2/kilogram	rectangular beads	\$3/kilogram	star-shaped beads	\$2/kilogram	spherical beads	\$2/kilogram	heart-shaped beads	\$3/kilogram	square beads	\$2/kilogram	flower-shaped beads	\$2/kilogram
oval beads	\$2/kilogram															
rectangular beads	\$3/kilogram															
star-shaped beads	\$2/kilogram															
spherical beads	\$2/kilogram															
heart-shaped beads	\$3/kilogram															
square beads	\$2/kilogram															
flower-shaped beads	\$2/kilogram															

- LLMs show strong text reasoning; true multimodal reasoning remains unresolved
- Many benchmarks allow language-only shortcuts via captions, weakening visual tests
- We need items where visual reasoning is **essential**, not ancillary or replaceable
- Prior work highlights this leakage and calls for stricter curation (Lu et al., 2024b; Wang et al., 2024a; Zhang et al., 2024a)

[Lu et al., 2024b; Wang et al., 2024a; Zhang et al., 2024a]

03 Introducing EMMA

Multimodal reasoning question in EMMA

Question: Three point charges, of charge $+Q$, $-2Q$, and $+3Q$, are placed equidistant as shown. Which vector best describes the net direction of the electric force acting on the $+Q$ charge?

Human reasoning

The direction of the electric force due to $-2Q$ and $+3Q$ can be visualized with a quick free-body sketch of the electric forces:

Combining the forces:
 $\because +3Q > |-2Q|$
 $\therefore \text{repel} > \text{attract}$
 b. ☒

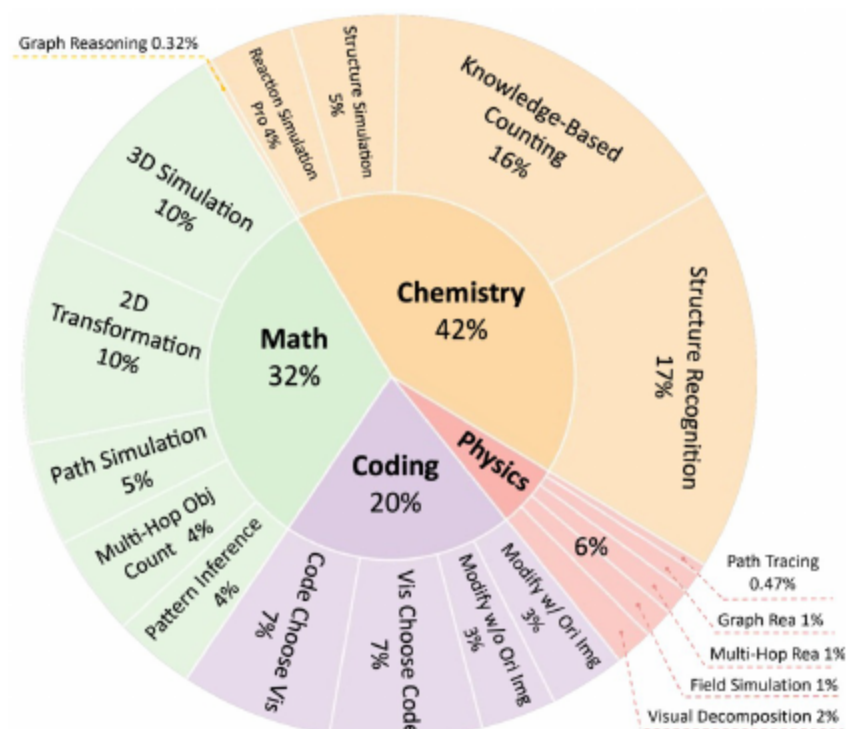
Model reasoning (GPT-4o)

- Forces Due to the Charges:**
 - The $-2Q$ charge attracts the $+Q$ charge. (Force points downward-left).
 - The $+3Q$ charge repels the $+Q$ charge. (Force points downward-right). **X**
- Force Magnitudes (Coulomb's Law):**
 - Force due to $-2Q$: $F_{-2Q} = \frac{26Q^2}{r^2}$.
 - Force due to $+3Q$: $F_{+3Q} = \frac{36Q^2}{r^2}$.

Error: While the model understands that like charges repel, it incorrectly identifies the force from $+3Q$ to $+Q$ as downward-right, when it should be upward-left.

- EMMA targets problems requiring both visual and textual reasoning steps
- Emphasizes back and forth interpretation across modalities, not one pass perception
- Avoids tasks solvable by text-only or a single glance, curbing **shortcut** solutions
- Outcome:** clearer assessment of multimodal reasoning, not just recognition

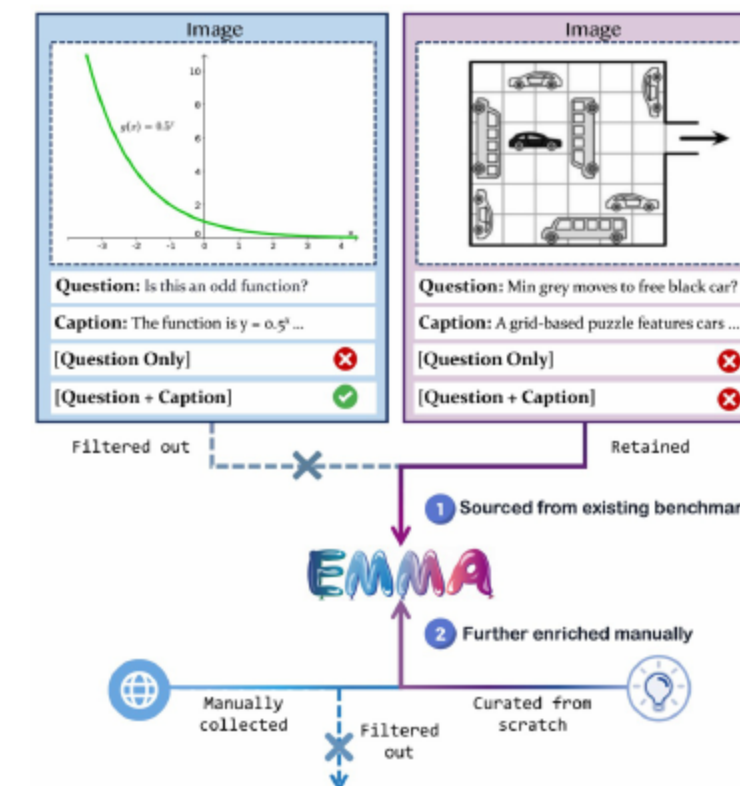
04 Domains and Skill Labels



- Four domains: **math**, **physics**, **chemistry**, **coding**
- Fine grained skill taxonomy enables targeted error analysis
- Two thousand seven hundred eighty eight questions with expert verified labels probe diverse multimodal skills
- Labels cover spatial simulation, graph reasoning, structure recognition, and visualization coding

05 Filtering Out Text-Only Shortcuts

- Two step filtering: text only, then text plus caption to remove **leakage**
- Discard any item if any model solves five or more of ten trials
- Ensures remaining items demand genuine visual reasoning beyond perception
- Stricter than prior pipelines, mitigating **caption based** shortcuts



[Yue et al., 2024a; Yue et al., 2024b]

06 Data, Human Baseline, Scope

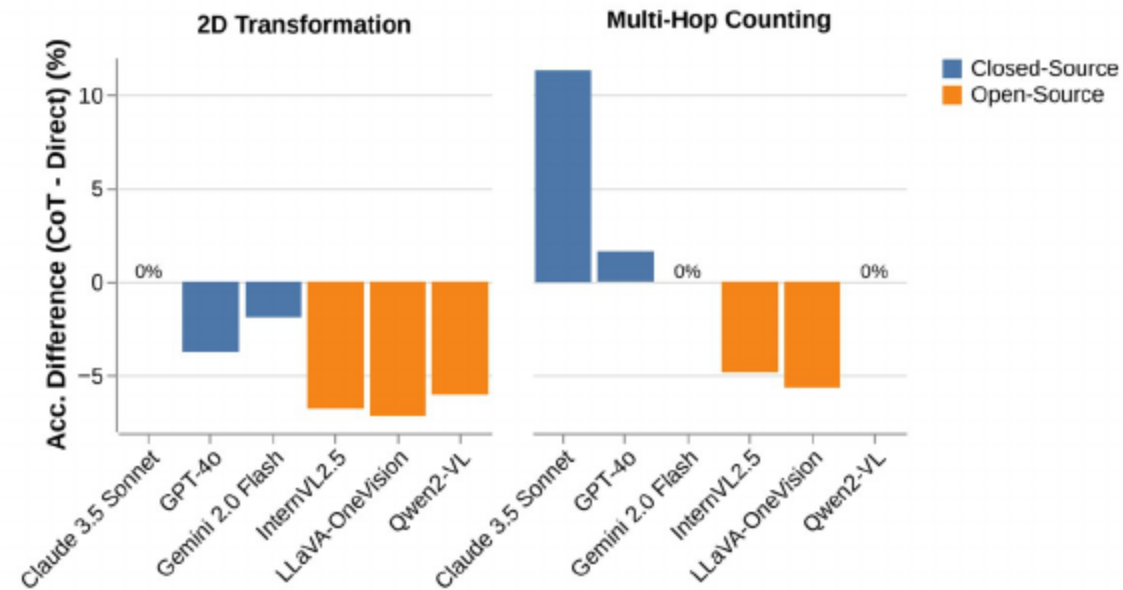
Statistic	Number
Total questions	2,788
- Multiple-choice questions	2,002 (72%)
- Free-form questions	786 (28%)
- Questions with answers	2,788 (100%)
- Questions newly added	1,796 (64%)
Image in the question	2,599 (93%)
Image in the option(s)	195 (7%)
Problems with multiple images	298 (10%)

- EMMA mini: four hundred balanced questions across subjects
- Human experts provide baseline performance for context
- Short answer multiple choice and open ended evaluation with clear ground truth
- Zero shot evaluation across ten state of the art multimodal models

07 Headline Accuracy and Human Gap

	CoT	EMMA					EMMA-mini				
		Math	Phys.	Chem.	Coding	Overall	Math	Phys.	Chem.	Coding	Overall
Random choice	—	(892)	(156)	(1,176)	(564)	(2,788)	(100)	(100)	(100)	(100)	(400)
Human Expert	—	14.01	25.64	16.50	25.71	18.08	75.00	64.50	86.00	85.50	77.75
Claude 3.5 Sonnet	✗	25.34	33.97	40.90	38.65	35.08	23.00	34.00	44.00	35.00	34.00
Gemini 2.0 Flash	✗	23.88	38.46	36.31	42.02	33.61	20.00	40.00	36.00	41.00	34.25
GPT-4o	✗	27.24	38.46	31.89	40.07	32.42	30.00	38.00	33.00	40.00	35.25
Qwen2-VL-72B-Instruct	✗	33.07	42.31	32.06	34.57	33.46	38.00	40.00	34.00	37.00	37.25
LLaVA-Onevision-72B	✗	27.69	35.90	25.26	28.72	27.33	25.00	32.00	24.00	28.00	27.25
InternVL2-Llama3-76B	✗	25.11	22.44	24.06	27.84	25.07	31.00	22.00	21.00	28.00	25.50
InternVL2.5-78B	✗	31.39	38.46	35.20	31.91	33.50	30.00	40.00	38.00	33.00	35.25
Claude 3.5 Sonnet	✓	29.37	41.03	41.07	40.60	37.23 (↑ 2.15)	30.00	38.00	41.00	39.00	37.00 (↑ 3.00)
Gemini 2.0 Flash	✓	25.90	38.46	24.66	40.96	29.12 (↓ 4.48)	24.00	41.00	36.00	44.00	36.25 (↑ 2.00)
GPT-4o	✓	25.56	43.59	33.67	39.01	32.71 (↑ 0.29)	27.00	44.00	35.00	38.00	36.00 (↑ 0.75)
Qwen2-VL-72B-Instruct	✓	27.69	34.62	24.57	29.43	27.12 (↓ 6.35)	35.00	34.00	32.00	23.00	31.00 (↓ 6.25)
LLaVA-Onevision-72B	✓	22.42	15.38	22.70	30.67	23.82 (↓ 3.52)	23.00	26.00	23.00	29.00	25.25 (↓ 2.00)
InternVL2-Llama3-76B	✓	22.20	32.05	19.73	30.32	23.35 (↓ 1.72)	27.00	33.00	21.00	32.00	28.25 (↑ 2.75)
InternVL2.5-78B	✓	25.56	39.74	27.47	25.18	27.08 (↓ 6.42)	31.00	36.00	24.00	19.00	27.50 (↓ 7.75)
Gemini 2.0 Flash Thinking-1219	—	31.61	56.41	37.93	43.44	38.06	35.00	57.00	41.00	41.00	43.50
Gemini 2.0 Flash Thinking-0121	—	37.11	60.26	41.58	48.05	42.50	34.00	63.00	47.00	48.00	48.00
QVQ-72B-Preview	—	—	—	—	—	—	34.00	39.00	24.00	31.00	32.00
o1	—	—	—	—	—	—	41.00	49.00	40.00	53.00	45.75

- Best model on EMMA mini reaches forty eight percent; humans score seventy seven point seventy five percent
- Closed source models generally outperform open source across subjects
- Large gaps persist in **math**, **chemistry**, and **coding**
- Physics gap is smaller but still **significant**



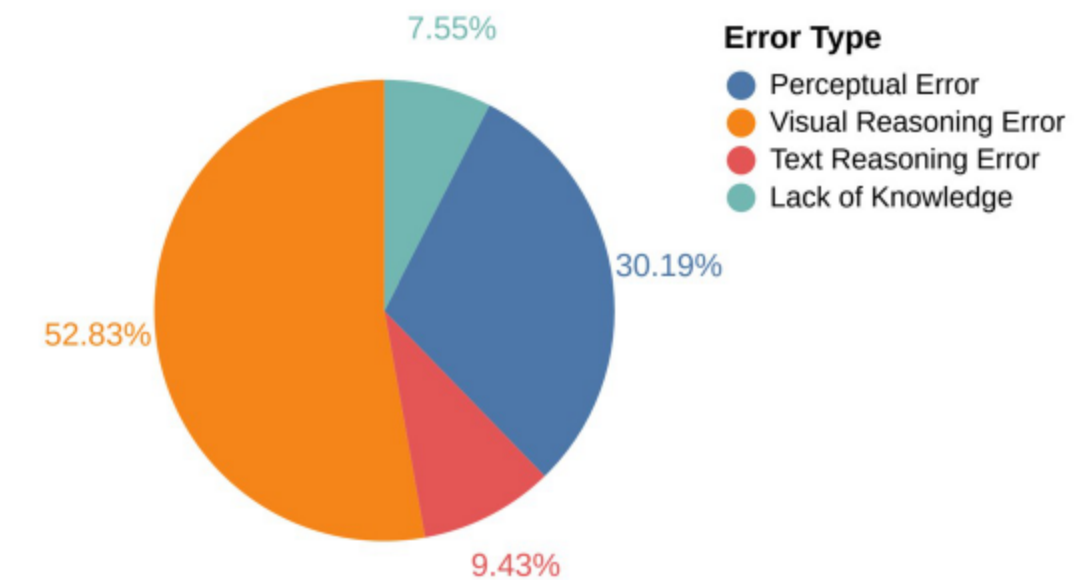
- Chain of thought helps many closed source models but often **hurts** open source
- Task dependence: visual heavy tasks benefit less from textual chain of thought
- Open source models show **more hallucinations** with chain of thought
- Implication:** language centric reasoning is not a substitute for visual simulation

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	—	—	—	37.25	36.25	38.25
	BoN	GPT-4o (Self)	—	35.50	35.75	36.75	—
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	—
	Tournament	Gemini Flash Thinking	—	40.75	39.25	41.25	35.25
	Pass@N	—	—	45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	—	—	—	37.75	39.25	39.75
	BoN	Gemini Flash (Self)	—	38.25	36.50	36.00	—
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	—
	Tournament	Gemini Flash Thinking	—	36.75	37.25	40.75	38.75
	Pass@N	—	—	45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	—	—	—	48.00	49.00	50.75
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	—	—	53.75	64.50	71.50	81.50
ol	—	—	45.75	—	—	—	—

- Evaluate majority voting, best of N, and tournament selection
- Reward models: self versus stronger external for response selection
- Generate candidates with chain of thought; experiment with N equal to one, two, four, eight, sixteen
- Goal:** quantify test time compute gains without retraining

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	—	—	—	37.25	36.25	38.25
	BoN	GPT-4o (Self)	—	35.50	35.75	36.75	—
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	—
	Tournament	Gemini Flash Thinking	—	40.75	39.25	41.25	35.25
	Pass@N	—	—	45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	—	—	—	37.75	39.25	39.75
	BoN	Gemini Flash (Self)	—	38.25	36.50	36.00	—
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	—
	Tournament	Gemini Flash Thinking	—	36.75	37.25	40.75	38.75
	Pass@N	—	—	45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	—	—	—	48.00	49.00	50.75
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	—	—	53.75	64.50	71.50	81.50
ol	—	—	45.75	—	—	—	—

- Improvements up to about seven and a half percent yet **far from** human level
- Stronger reward models beat self reward; tournament exceeds best of N
- Benefits saturate for weaker bases; stronger bases raise pass at N
- Takeaway:** scaling helps but does not close the multimodal reasoning gap

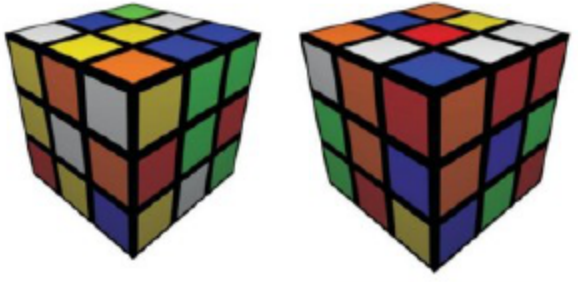


- Visual reasoning errors **dominate** overall failures
- Perceptual and textual errors are smaller fractions of mistakes
- Multi hop spatial simulation remains a recurring weakness
- Implication:** better visual imagination and manipulation are needed

12 Relation to Prior Benchmarks

▷ 3D Spatial Simulation

Image:



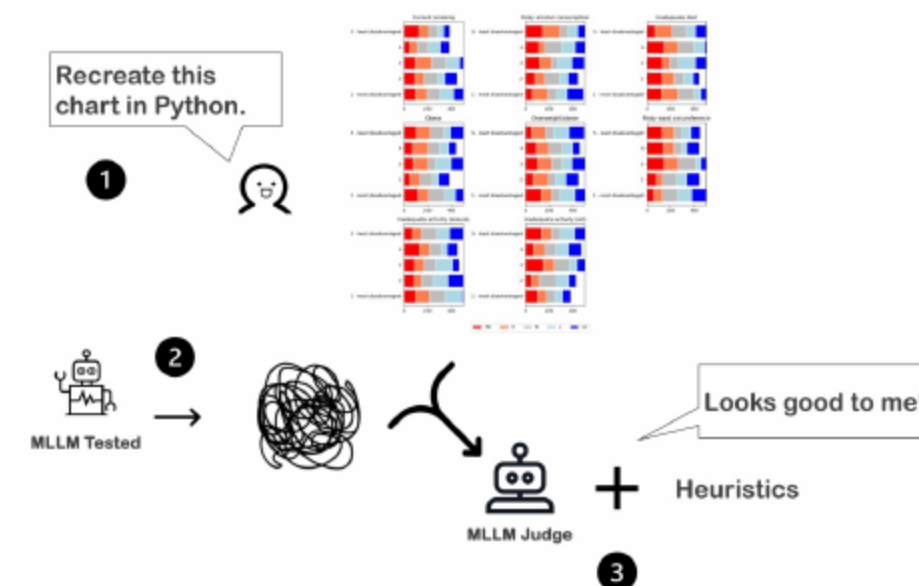
<image1>: <image2>:

Question: How many steps from the solution based on <image 1> and <image 2>?

- EMMA removes **caption based** shortcuts to enforce visual reasoning
- Tasks are mapped to skill categories for cross dataset comparability
- Focus on problems unsolvable by text only or a single visual pass
- Net effect:** stronger, more diagnostic multimodal assessment

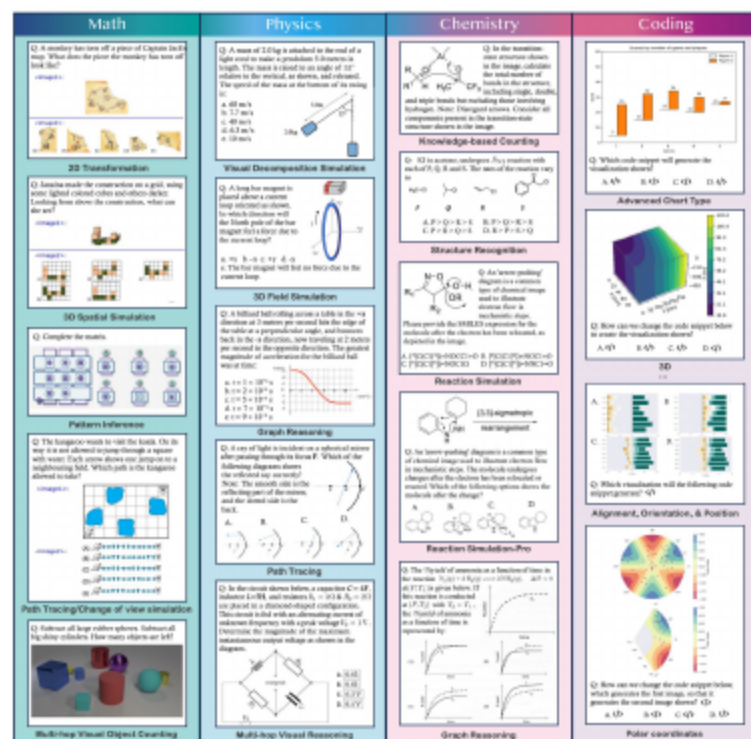
[Yue et al., 2024a]

13 Coding Task Design



- Multiple choice visualization tasks remove dependence on MLLM judges
- Fine grained categories enable targeted analysis of limitations
- Covers vision to code, code to vision, and modification workflows
- Benefit:** **reliable** scoring and broader skill coverage

14 Conclusion and Impact



- A **large** human model gap persists on multimodal reasoning
- Visual reasoning remains the primary bottleneck across domains
- Progress needs new **architectures, training, and evaluation** paradigms
- EMMA provides a **clear, diagnostics oriented** benchmark for future work

[https://emma-benchmark.github.io/]

THANKS!